

Communication Protocols

UART • SPI • I²C • CAN • RS-485 • USB • Ethernet

How embedded systems talk — wired & wireless

Audience: Beginner → Intermediate Engineering Students

Wiring • Speed • Topology • Use case • Debug

Why Protocols Matter

Every embedded system communicates

MCU ↔ peripheral

Sensors, displays, memory.

Board ↔ board

Sub-modules in a product.

Product ↔ host

PC, mobile, server.

Product ↔ product

Cars, factories, IoT mesh.

Product ↔ cloud

Updates, telemetry, OTA.

Debug ↔ developer

JTAG, SWD, console.

Protocol Cheatsheet

At-a-glance comparison

Protocol	Pins	Speed	Topology	Typical use
UART	2 (TX/RX)	≤ 3 Mbps	Point-to-point	Debug, GPS, modem
SPI	4+ (CS per slave)	≤ 50 MHz	1 master, N slaves	Sensors, flash, displays
I ² C	2 (SDA/SCL)	100k-3.4M	Multi-master/slave	Sensors, EEPROM, RTC
RS-485	2 (A/B)	≤ 50 Mbps	Multi-drop	Industrial, long cable
CAN	2 (H/L)	1-5 Mbps	Multi-drop bus	Automotive, industrial
USB 2.0	4 (D+/D-, V/G)	12-480 Mbps	Host/device	PC peripherals
USB 3.0+	8+ (diff pairs)	5-40 Gbps	Host/device	Storage, video
Ethernet	4-8 (diff)	10M-100G	Switched LAN	Networking
1-Wire	1	16 kbps	Star/daisy	DS18B20, ID chips
LIN	1	20 kbps	Single master	Automotive sub-bus

UART — Universal Asynchronous Receiver/Transmitter

The simplest serial protocol

- ▶ Two wires + ground: TX and RX
- ▶ Asynchronous — no clock — both ends agree on baud rate
- ▶ Common: 9600, 115200 baud; up to 3 Mbps
- ▶ Frame: start bit + 8 data + (parity) + stop bit
- ▶ TTL UART (3.3/5V), RS-232 ($\pm 12\text{V}$), RS-485 (differential)
- ▶ Most ubiquitous debug interface in firmware

Watch outs

Both ends must use exactly the same baud. $\pm 3\%$ mismatch breaks comms. Cross TX $\leftrightarrow$ RX on connection (TX of one to RX of other).

Practical use

GPS modules, modems, console (Linux `ttyS0/ttyUSB0`), Bluetooth modules, satellite comms. CP2102 / FT232 USB-UART bridges are essential bench tools.

SPI — Serial Peripheral Interface

Fast, full-duplex, master-slave

- ▶ Four wires: MOSI, MISO, SCK, CS
- ▶ Master drives SCK; data shifted in/out on each clock edge
- ▶ Mode 0/1/2/3 selects clock polarity (CPOL) and phase (CPHA)
- ▶ Multiple slaves via one CS line each — no addresses on the bus
- ▶ Speeds 1 MHz to 50+ MHz; full-duplex (read while writing)
- ▶ Used by: SD card, OLED display, flash memory, IMU, high-speed ADC
- ▶ Pin-cost grows with slave count — limits density
- ▶ Daisy-chain mode possible but rare

I²C — Inter-Integrated Circuit

Two wires, many devices, short distance

- ▶ SDA + SCL, both open-drain with pull-ups
- ▶ Each slave has a 7-bit (or 10-bit) address
- ▶ Standard 100 k, Fast 400 k, Fast+ 1 M, High-speed 3.4 M
- ▶ Multi-master capable (rare in practice)
- ▶ Clock stretching — slave holds SCL low to slow master
- ▶ Common bus length ~ 1 m at 100 k; less at higher speed

Pull-up sizing

4.7 kΩ standard for 100 k, 2.2 kΩ for 400 k, 1 kΩ for 1 M. Wrong pull-up = no comms or marginal at temperature.

Bus lock-ups

If a slave holds SDA low, the bus is stuck. Have a 'reset slave' GPIO or power-cycle option. SMBus has timeout — I²C does not.

SPI vs I²C

Choosing between the two

Choose SPI when

- ▶ High data rate needed
- ▶ Few slaves, plenty of pins
- ▶ Full-duplex needed
- ▶ Long cable / noisy environment
- ▶ No address conflicts (each CS)

Choose I²C when

- ▶ Many devices on the same bus
- ▶ Pin count budget is tight (2 vs 4+)
- ▶ Slow-to-medium speed acceptable
- ▶ On-board, short connections
- ▶ Devices have built-in addressing

RS-485 — Industrial Long-Distance

Differential UART for factories

- ▶ Two-wire differential pair (A, B) + ground
- ▶ Up to 1.2 km at 100 kbps; 50 Mbps at short range
- ▶ Up to 32 devices on the same bus (or 256+ with repeaters)
- ▶ Multi-drop topology — needs termination at both ends (120 Ω)
- ▶ Industrial protocols on top: Modbus RTU, Profibus, DMX (lighting)
- ▶ Half-duplex (most common) or full-duplex (4-wire)
- ▶ Failsafe biasing required: pull A high, B low
- ▶ Transceivers: MAX485, SN65HVD75, ISL4485 (isolated)

CAN Bus — Automotive Industrial

Robust, arbitration, broadcast

- ▶ Two-wire differential (CAN_H, CAN_L) + ground
- ▶ Up to 1 Mbps (Classical) or 5 Mbps (CAN-FD)
- ▶ Multi-master with arbitration — lowest ID wins
- ▶ Message-based (not address-based) — every node sees every frame
- ▶ 120 Ω termination at both ends of a linear bus
- ▶ Used by: cars (every modern car has ≥ 3 CAN buses), industrial PLCs, marine
- ▶ Protocols on top: J1939, CANopen, OBD-II, DeviceNet
- ▶ Transceivers: TCAN334, SN65HVD230

USB — The Universal Bus

From keyboards to thunderbolt

Speed	Class	Wires	Use
Low Speed	USB 1.0	1.5 Mbps	Keyboards, mice
Full Speed	USB 1.1	12 Mbps	Cheap MCU USB
High Speed	USB 2.0	480 Mbps	Storage, audio
Super Speed	USB 3.0	5 Gbps	SSD, cameras
USB 3.2	USB 3.2	10/20 Gbps	Modern storage
USB4 / TB4	USB4	40 Gbps	Docking stations, displays

USB-C is a connector, not a speed

USB-C the connector can carry USB 2.0, USB 3.x, USB4, Thunderbolt, DisplayPort, and PD (up to 240 W) on the same physical port.

Ethernet

From 10BASE-T to 100GbE

- ▶ 10/100 Mbps — common for industrial PLCs, IoT gateways
- ▶ 1 Gbps — server, modern industrial
- ▶ 10 Gbps+ — datacenter, AI cluster
- ▶ MAC (in MCU/SoC) ↔ PHY (external IC) ↔ Magnetics ↔ RJ45 ↔ cable
- ▶ Common PHYs: LAN8720 (10/100), KSZ8081 (10/100), DP83867 (gigabit)
- ▶ Layer-2: MAC addresses, switch learning
- ▶ Layer-3: IP — Internet Protocol, routing
- ▶ Industrial Ethernet variants: EtherCAT, PROFINET, EtherNet/IP

Less-Common but Useful

Protocols you'll meet in special domains

1-Wire

Single signal + GND.
DS18B20 thermometer.

LIN bus

12V automotive single-wire.
Window motors, climate.

Manchester

Self-clocking — RFID,
PLC remote heads.

MIPI CSI

Camera serial interface.
Differential, 1-4 GHz.

MIPI DSI

Display serial interface.
Phones, tablets.

PCIe

Computer expansion bus.
NVMe storage, GPUs.

Wireless Protocols — Quick Tour

When wires can't reach

Protocol	Range	Data rate	Best use
Bluetooth LE	10-50 m	1-2 Mbps	Wearable, sensor, audio
Wi-Fi 4/5/6	30-100 m	100-1000 Mbps	Consumer IoT, gateway
Zigbee	10-100 m	250 kbps	Mesh sensor networks
LoRa / LoRaWAN	2-15 km	0.3-50 kbps	Outdoor IoT, low data
NB-IoT / LTE-M	cellular	20-100 kbps	Wide-area IoT
UWB	10-30 m	100 Mbps	Indoor ranging, AirTag-style
NFC	≤ 5 cm	424 kbps	Tap-to-pay, pairing

Debugging Serial Buses

Logic analyzer is your best friend

Logic analyzer

Saleae Logic, Kingst,
LA1010 (DSLogic).

Protocol decoder

Most logic analyzers
decode UART/SPI/I²C live.

Oscilloscope

See edges, ringing,
levels, glitches.

Bus pirate

Hand-craft transactions
to sensor.

USB sniffer

Wireshark + USBPcap
for USB analysis.

CAN tool

PEAK, Kvaser, USB2CAN
to observe automotive bus.

Common Protocol Mistakes

What logic-analyzer captures reveal

- ▶ Missing pull-ups on I²C — no acks, comms fail intermittently
- ▶ Wrong SPI mode (CPOL/CPHA) — bytes come back garbled
- ▶ UART baud mismatch — random characters or none
- ▶ RS-485 bus without 120 Ω termination — reflections, errors
- ▶ CAN without 120 Ω at both ends — same issue
- ▶ USB without USB-spec'd connectors / cables — works for some hosts only
- ▶ Long I²C cable trying 400 kHz — drop to 100 k or use level-shifters
- ▶ Floating CS line on SPI slave — random selection by noise

Best Practices

Habits of senior firmware engineers

- ▶ Pick the simplest protocol that meets your needs
 - ▶ Document baud / pinout in the schematic
 - ▶ Add test points for every bus line
 - ▶ Use DMA for high-speed UART / SPI / I²C
 - ▶ Add CRC on every frame in noisy environments
 - ▶ Use a logic analyzer at first bring-up
- ▶ Plan for protocol failures (timeouts, retries)
 - ▶ Watchdog timer that monitors comms
 - ▶ Bootloader & DFU for OTA firmware updates
 - ▶ Pull-up / pull-down on every floating input
 - ▶ ESD / TVS protection on every external pin
 - ▶ Version-tag your protocols (v1, v2) for upgrades

Key Takeaways

Key takeaways from this lecture

Right protocol per job

Speed + distance + pin budget.

I²C ≠ SPI

Multi-drop vs single fast peripheral.

Long distance = differential

RS-485, CAN beat single-ended.

USB-C is a connector

Many protocols share the port.

Logic analyzer essential

Debug in seconds, not days.

Protect every external pin

TVS + ESD = compliance-ready.

Questions & Discussion

Ask anything — concepts, projects, sourcing, career.

References & Further Learning

- ▶ Jack Ganssle — embedded protocols lectures (free)
- ▶ Saleae Logic — free trial of analyzer software
- ▶ Wireshark — packet analyzer for USB/Ethernet
- ▶ ISO 11898 — CAN standard (free abstract)
- ▶ USB-IF specifications (free)
- ▶ Bosch CAN tutorial (free PDF)
- ▶ Maxim Integrated Application Notes — 1-Wire / I²C
- ▶ RS-485 design notes from MaxLinear / Linear Technology

Thank you • Build something this week.